Perceptual Quality Assessment of Low-light Image Enhancement

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Low-light image enhancement algorithms (LIEA) can light up images captured in dark or back-lighting conditions. However, LIEA may introduce various distortions such as structure damage, color shift, and noise into the enhanced images. Despite various LIEAs proposed in the literature, few efforts have been made to study the quality evaluation of low-light enhancement. In this article, we make one of the first attempts to investigate the quality assessment problem of low-light image enhancement. To facilitate the study of objective image quality assessment (IQA), we first build a large-scale low-light image enhancement quality (LIEQ) database. The LIEQ database includes 1,000 light-enhanced images, which are generated from 100 low-light images using 10 LIEAs. Rather than evaluating the quality of light-enhanced images directly, which is more difficult, we propose to use the multi-exposure fused (MEF) image and stack-based high dynamic range (HDR) image as a reference and evaluate the quality of low-light enhancement following a full-reference (FR) quality assessment routine. We observe that distortions introduced in low-light enhancement are significantly different from distortions considered in traditional image IQA databases that are well-studied, and the current state-of-the-art FR IQA models are also not suitable for evaluating their quality. Therefore, we propose a new FR low-light image enhancement quality assessment (LIEQA) index by evaluating the image quality from four aspects: luminance enhancement, color rendition, noise evaluation, and structure preserving, which have captured the most key aspects of low-light enhancement. Experimental results on the LIEQ database show that the proposed LIEQA index outperforms the state-of-the-art FR IQA models. LIEQA can act as an evaluator for various low-light enhancement algorithms and systems. To the best of our knowledge, this article is the first of its kind comprehensive low-light image enhancement quality assessment study.

CCS Concepts: • **Computing methodologies → Model development and analysis;**

Additional Key Words and Phrases: Low-light image enhancement, quality assessment, light-enhanced image, structure similarity

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1 INTRODUCTION

Images captured by the visible light camera may suffer from degradations such as low visibility, low contrast, and strong noise in the low or back-lighting environment. Such degraded images will make people feel unsatisfactory and also challenge many computer vision algorithms [91] such as image classification, image segmentation, object tracking, and so on, which are usually designed for high-visibility images. Though we can alleviate such degradations to some extent through using the professional equipment and software, it requires users to have professional photographic skills and can not be used in applications that need real-time processing. Therefore, single **low-light image enhancement algorithms (LIEA)** have been developed [9, 11–13, 18, 20, 29, 73, 82, 83] to lighten low-light images automatically.

1.1 Low-light Image Enhancement Algorithms

Many low-light image enhancement algorithms have been proposed in recent years. In general, LIEAs can be classified into histogram equalization-based, retinex theory-based, and learning-based methods. We shortly review several representative LIEAs for each category.

Histogram equalization-based methods stretch the dynamic range of low-light images to the full pixel range and avoid saturation in some relatively bright areas. **Histogram equalization (HE)** directly maps the gray levels of low-light images based on the probability distribution of the input gray levels. Ibrahim et al. [20] introduce brightness-preserving dynamic histogram equalization for image contrast enhancement. Celik et al. propose a **Contextual and Variational Contrast (CVC)** enhancement method [6] through considering inter-pixel contextual information. Lee et al. develop a contrast enhancement algorithm based on layered difference representation of 2D histogram [28]. However, the purpose of HE-based methods is primarily to increase image contrast. These methods pay more attention to the global information of the image while ignoring the local structure, which often causes over- and under-enhancement.

Retinex theory has also been widely used to enhance low-light images. Retinex theory assumes that the color image can be decomposed into reflectance and illumination components, which can be expressed by

$$\mathbf{I} = \mathbf{R} \circ \mathbf{L}, \tag{1}$$

where $\mathbf{I}$ is the captured image, and $\mathbf{R}$ and $\mathbf{L}$ represent the reflection and illumination components of the image, respectively. The operator $\circ$ means element-wise multiplication. Early studies such as single-scale retinex [24] and multi-scale retinex [23] directly regard $\mathbf{R}$ as the enhanced result, which often causes enhanced images to look unnatural due to the lack of luminance component. To preserve naturalness while enhancing details, Wang et al. [73] first use a bright-pass filter to decompose the observed image into reflection and illuminance and then utilize a bi-log transformation to increase the illuminance while preserving naturalness. Fu et al. [13] propose a weighted variational model to estimate the reflection and illuminance simultaneously by imposing the prior representation in the regularization terms, which can compensate for the defects caused by the logarithmic transformation. They [11] further estimate illuminance and reflectance in the linear domain, and show that the linear domain can better represent prior information than the logarithmic domain. Guo et al. [18] propose a structure-aware smoothing model to estimate the illuminance component, which can shrink the solution space and reduce the computational cost. Li et al. [29] develop a robust retinex model to estimate a piece-wise smoothed illuminance and a structure-revealed reflectance through considering intensive noise term. Ying et al. [83] first develop an accurate camera response model and utilize the model to adjust pixels of the input image to its perfect exposure according to the estimated exposure ratio map. The image fusion technology is also introduced to enhance low-light images. For example, Fu et al. [12] fuse multiple derivatives of the estimated illumination to blend the advantages of them into a single output. Ying et al. [82] use the camera response model to synthesize the well-exposed image and then fuse the well-exposed image with the original image based on the illuminance weight matrix.

Recently, the deep **convolution neural network (CNN)** shows a great performance for image enhancement. Lore et al. [33] propose a deep autoencoder approach to simultaneously perform contrast enhancement and denoising. Shen et al. [64] show that **multi-scale retinex (MSR)** is equivalent to a feedforward CNN with different Gaussian convolution kernels and then propose the MSR-net, which consists of multi-scale logarithmic transformation, difference-of-convolution, and color restoration function to learn an end-to-end mapping from dark to bright images. Lv et al. [34] utilize multiple subnets to extract features from different levels and then merge feature maps with different levels into the enhanced image by multi-branch fusion. Cai et al. [4] design a deep image contrast enhancer (SCIE) to enhance the low-light image from the low-frequency luminance component and the high-frequency detail component and then merge them into final enhanced results. Wang et al. [72] propose a network to first estimate an image-to-illuminance mapping for modeling varying-lighting environments and fuse the illuminance map with the original image to enhance the underexposed image.

1.2 Quality Assessment of Low-light Enhancement

Image quality assessment (IQA) is very important to promote the development of low-light enhancement methods. On the one hand, when we propose a new LIEA, we need to compare it with other state-of-the-art methods. On the other hand, when we apply the LIEAs, we need to select the best algorithm. Generally, image quality can be evaluated using two strategies: subjective [67, 68] and objective [44, 46, 85, 86]. In the literature, both subjective [72] and objective [12, 18, 83] quality assessment are used to assess the quality of light-enhanced images. Subjective quality assessment is based on the perceptual quality ratings of human subjects. It is more straightforward and accurate, since humans are usually the ultimate receivers. However, subjective quality assessment is time-consuming and expensive. So it can not be used for large-scale enhanced image evaluation and also can not be embedded in the real-time processing systems. Considering such drawbacks, objective quality assessment is introduced to predict perceptual quality automatically.

Generally speaking, objective IQA algorithms [84] can be classified into **full-reference (FR), reduced-reference (RR)**, and **no-reference (NR)** [17, 41, 42, 69] according to the existence of the reference image. FR IQA and RR IQA require full and partial reference image information, respectively, while NR IQA only takes the distortion image as input. In this article, we focus on the FR quality assessment. Most FR IQA models are proposed for evaluating images distorted by some common distortion types such as JPEG compression, Gaussian noise, blur, and so on. For instance, PSNR and **Structure Similarity Index (SSIM)** [74] are the two most widely used FR IQA methods. PSNR calculates the pixel-to-pixel difference between the distorted and reference images, and SSIM calculates luminance, contrast, and structure similarity between two images. **Most Apparent Distortion (MAD)** [27] index first uses contrast sensitivity, local luminance, and contrast masking to measure high-quality images, then uses the changes in local statistics between two images to measure low-quality images, and finally combines the two measures. **Visual**

saliency-induced (VSI) [87] index considers visual saliency [45] as the important quality features and then calculates the similarity of visual saliency, gradient magnitude, and chrominance as the image quality. **Feature similarity (FSIM)** [88] index utilizes the **phase congruency (PC)** and the image gradient magnitude to determine the local image quality, then the PC is also used to pool the local quality. **Gradient Magnitude Similarity Deviation (GMSD)** [78] index calculates the pixel-wise **gradient magnitude similarity (GMS)** as the local quality map and then pools it with the standard deviation of GMS. Zhang et al. [89] propose that using deep features of the pre-trained **deep neural network (DNN)**, as the perceptual metric can be more consistent with human visual system.

These metrics have proven to be very effective in evaluating image quality on the traditional IQA database such as LIVE [74], CSIQ [27], TID2008 [56], and TID2013 [55], which consist of simulated distorted types such as JPEG compression, Gaussian noise, blur, and so on. However, the distortions introduced by low-light enhancement are much more complicated than these simulated distortions. Various kinds of distortion types such as structure damage, color shift, noise, over-enhancement, and so on, may be introduced in the process of low-light image enhancement. What is more, light-enhanced images may contain one or more kinds of these distortions, which makes them more different and also much more complicated than distorted images in the commonly used IQA database. Therefore, state-of-the-art FR IQA algorithms are not suitable for low-light image enhancement quality evaluation, which will be verified in this article.

Besides the traditional IQA algorithms, there are also some IQA algorithms specifically designed for enhanced images, such as super-resolution images [79], dehazing images [43], tone-mapped images [51, 81] and so on. Among them, the most similar to our research is the tone-mapping IQA. **Tone-mapping operators (TMO)** [10, 38, 58] are used to map **high dynamic (HDR)** images to **low dynamic range (LDR)** images while preserving the image details and color appearance as much as possible so HDR images can be rendered on standard LDR displays. But due to the strong contrast reduction, TMOs may introduce distortions such as under-exposure, over-exposure, color shift, halos, over-enhancement ("cartoon-like" appearance), and so on, which have certain degree of overlap with the distortions caused by LIEAs. To evaluate the visual quality of tone-mapped images generated by different TMOs, many quality metrics are proposed [51, 81]. For example, TMQI [81] evaluates the quality of tone-mapped images by combining a multi-scale signal fidelity measure modified from SSIM and a naturalness measure. FSITM [51] is a feature similarity index for tone-mapped images, which compares the locally weighted mean phase angle maps of the HDR and LDR images. Kundu et al. [26] designed a NR tone-mapping IQA using space-domain NSS features and HDR-specific gradient-based features. BLIQUE-TMI [22] assesses the quality of tone-mapped images from three aspects: visual information, local structure, and naturalness. Mahmoudpour and Schelkens [37] developed an NR IQA model for tone-mapped images based on multi-attribute features, such as spectral and spatial entropy, detection probability of visual information, image exposure, and so on.

In the previous studies of low-light image enhancement, some FR IQA measures such as PSNR, SSIM, GMSD, FSIM, or some NR IQA measures such as NIQE [49], NFERM [17] are frequently used to evaluate the enhancement effect. However, in this article, we show that these traditional IQA methods have a low correlation with the subjective perceptual quality. The study in Reference [73] proposes a **lightness order error (LOE)** method to objectively measure the lightness distortion of enhanced results. Some papers calculate the color distortion [83], contrast gain [11], discrete entropy [73] to evaluate the quality of enhanced images in terms of color shift, contrast enhancement, and randomness. But these descriptors only measure the quality from a certain and limited aspect, rather than the overall enhancement quality. They are insufficient to assess the quality of low-light enhancement.

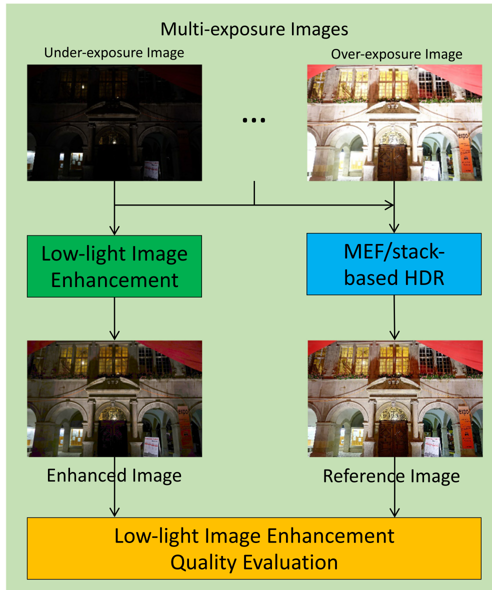


Fig. 1. Framework of the low-light image enhancement quality evaluation.

1.3 The Contribution of This Article

In this article, we study both subjective and objective low-light image enhancement quality assessment comprehensively. The subjective study can provide a benchmark for state-of-the-art LIEAs, while the developed objective quality assessment method can act as a quality evaluator for low-light enhancement. To facilitate the research, we first build a large **low-light image enhancement quality (LIEQ)** database, which includes 1,000 enhanced images derived from 100 low-light images using 10 low-light image enhancement methods. The low-light images are selected from a multi-exposure image dataset [4], which consists of multi-exposure image sequences and their corresponding well-exposed reference ones generated by a group of MEF algorithms and stack-based HDR algorithms. We select low-light images with various exposure levels to make our database more diverse and comprehensive. A subjective quality assessment study is then conducted on the LIEQ database. The **mean opinion scores (MOSs)** are collected as the ground-truth qualities of light-enhanced images.

Then, we propose a **low-light image enhancement quality assessment (LIEQA)**. A framework of this low-light image enhancement quality assessment study is illustrated in Figure 1. Considering that evaluating the quality of the light-enhanced image is more difficult, we propose to use the multi-exposure fused image and stack-based HDR image as a reference and evaluate the quality of low-light enhancement following a **full-reference (FR)** IQA framework. Such kind of reference has acted as the ground-truth of low-light enhancement in many current works. The low-light enhancement quality is evaluated from four aspects: luminance enhancement, color rendition, noise evaluation, and structure preserving.

Specifically, luminance enhancement is measured as the **earth mover's distance (EMD)** [60] between the reference and light-enhanced images. We further normalize it by dividing the EMD between the reference image and a black image to eliminate the influence of original image content luminance. The color rendition is evaluated as the product of the hue and saturation similarity map. The **local binary pattern (LBP)** [53] is used to measure the intrinsic noise amplified by LIEAs.

The structure preserving is evaluated from two aspects: structure damage and over-enhancement, which are measured by the local variance similarity and edge structural similarity, respectively. Finally, we integrate these four components as the overall quality of the light-enhanced image. We verify the performance of the proposed method on the LIEQ database. The results show that the proposed method is significantly superior to the current state-of-the-art FR IQAs and other quality descriptors frequently used in LIEAs studies. To the best of our knowledge, both the LIEQ database and the LIEQA index are first-of-its-kind. The LIEQ database is the largest low-light image enhancement quality database so far, while the LIEQA index is the first comprehensive low-light enhancement quality evaluator.

The rest of this article is organized as follows: Section 2 introduces the construction of the light-enhanced image quality database and the subjective user study. In Section 3, we describe the proposed LIEQA algorithm in detail. Experimental verification is given in Section 4. Section 5 gives the concluding remarks.

2 SUBJECTIVE QUALITY ASSESSMENT OF LOW-LIGHT IMAGE ENHANCEMENT

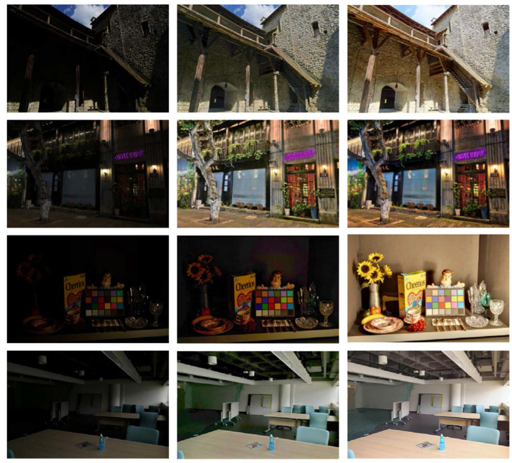
In this section, we first build a large-scale LIEQ database. A subjective quality assessment study is subsequently conducted to obtain the ground-truth qualities of enhanced images. This subjective study can not only provide a large-scale subjective benchmark for the state-of-the-art LIEAs, but also facilitate the following objective low-light enhancement quality evaluation study.

2.1 The Low-light Image Enhancement Quality Database

To construct a low-light image enhancement quality database, it will be better if we can have both low-light images and their corresponding well-light ones. In practice, it is difficult to capture the low-light image and its well-light image simultaneously. Some studies propose to use a synthesized low-light image derived from a well-light image or tone-map the low-light image to a good one using professional software. However, both methods have their drawbacks. The synthesized low-light image is usually simulated by using gamma correction or histogram specification methods. It is difficult to simulate the unbalance illuminance and intrinsic noise captured in the dark environment. Tone-mapping can not recover the image structure that the real low-light images do not have.

In this article, we select test images from the multi-exposure image database in Reference [4], which resorts to **multi-exposure fusion (MEF)** and stack-based HDR techniques to solve this problem. Multi-exposure fusion aims to merge the same scene with multiple exposures into a single well-exposed image, while stacked-based HDR aims to merge multiple exposure images into an HDR irradiance map and then implement a tone-mapping operator to convert the HDR image to the LDR image. To derive the high-quality reference images, the authors in Reference [4] first used 13 state-of-the-art MEF and stack-based HDR algorithms to generate candidate reference images. Specifically, there are eight MEF methods: Mertens09 [39], Raman09 [57], Shen11 [65], Zhang12 [90], Li13 [31], Shen14 [63], Ma17 [36], Kou17 [25], and five stack-based HDR methods: Sen12 [61], Hu13 [19], Bruce14 [2], Oh15 [52], Photomatix [54]. Then 18 volunteers participated in the subjective experiment to choose the best reference image of each scene from the candidate images. For each volunteer, the best result of a scene will be selected after 12 pairs of comparisons, and then the best reference image of each scene is selected by the majority vote of all volunteers. Further, for those challenging multi-exposure images that contain unaligned contents, even the best MEF/HDR results can not have satisfactory quality, and these unsatisfactory images are abandoned from the database. Therefore, all reference images in the database are of high quality.

We select 100 under-exposure images that have various lighting conditions and their reference ones from the above multi-exposure image database. Ten representative low-light image



(a) (b) (c)

Fig. 2. Example images of the LIEQ database: (a), (b), and (c) are the low-light images, enhanced images, and reference images, respectively.

enhancement algorithms, including histogram equalization (HE), BPDHE [20], NPEA [73], Dong [9], PLE [11], SIRE [13], FEM [12], LIME [18], CRM [83], and EFF [82] are employed to enhance low-light images. A total of 1,000 enhanced images are generated. All enhanced images and their corresponding low-light, well-light images together constitute the LIEQ database. Some sample images of this database are illustrated in Figure 2.

2.2 Subjective Quality Evaluation Study

We perform a subjective quality assessment study on the LIEQ database. The general methodology and configuration of the subjective test are presented as follows:

- **Participants:** A total of 21 subjects recruited from the university participate in the experiment. The ages of participants range from 20 to 29. All of them have the normal or corrected-to-normal vision.
- **Method:** All test images are displayed in a random order using a MATLAB graphical user interface on a LCD monitor with a resolution of 1,980 × 1,200, which is calibrated according to the recommendations of ITU-R BT.500-13 [3]. Both the low-light image and the reference image are shown on the monitor [14] to help the subject perceive and evaluate the quality of the enhanced image. All images are displayed at the original resolutions. To avoid fatigue, the whole test is divided into three sessions, and each session lasts less than 30 minutes.
- **Test Condition:** The experiments are conducted in an empty room with a normal indoor illuminance level and no noise. The subjects are seated at a viewing distance of around three times the image height.
- **Quality Rating:** Subjects are required to rate the overall enhancement quality with a five-grades continuous rating scale, where a higher value means better quality. The subjects are instructed to give the overall quality from three aspects: the enhanced image should be lighted up enough, should preserve the image content, and should not introduce new artifacts.

Before the formal experiment, clear instructions are made to each subject. Then subjects participate in a training session, in which they are instructed to view some example images that would

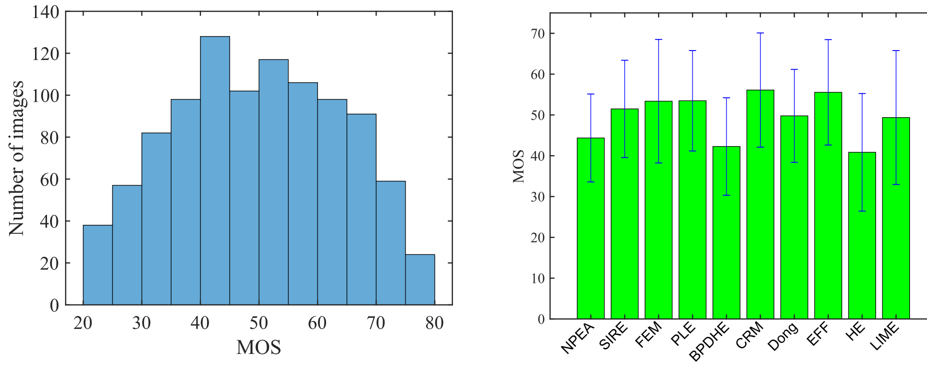


Fig. 3. Data analysis for LIEQ database: (a) histogram of MOSs in the LIEQ database, (b) the mean and standard deviation of subjective scores for all LIEAs in the LIEQ database.

not show in the formal experiment and give a proper rating. Through the training, the subjects can get familiar with the contents, distortions, GUI, and so on, and have an idea on how to provide their perceptual rating for the image quality. In the formal experiment, subjects view the images displayed on the monitor and then provide their perceptual ratings. After the subjective experiment, we collect the perceptual ratings and conduct further analysis.

2.3 Subjective Data Processing and Analysis

We follow the recommended method in Reference [3] to process the raw subjective ratings. First, the rating of an image is considered as an outlier if it is far from 2 (if normal) or $\sqrt{20}$ (if non-normal) standard deviations about the mean rating of that image. A subject with more than 5% outlier ratings is also rejected. One subject is rejected in our experiment. Then the **mean opinion score (MOS)** is calculated for each enhanced image. Let s_{ij} represent the raw rating assigned subject i to image j. We convert the score s_{ij} to a Z-score

$$Z_{ij} = \frac{s_{ij} - \mu_i}{\sigma_i}, \tag{2}$$

$$\mu_i = \frac{1}{N_i} \sum_{j=1}^{N_i} s_{ij}, \quad \sigma_i = \sqrt{\frac{1}{N_i - 1} \sum_{j=1}^{N_i} (s_{ij} - \mu_i)^2}, \tag{3}$$

where N_i denotes the number of test image viewed by subject i. Then Z-score Z_{ij} is linearly rescaled to lie in the range of $[0, 100]$ and the MOS of the image j is calculated by averaging the Z-score of image j from M_j subjects

$$Z'_{ij} = \frac{100(Z_{ij} + 3)}{6}, \quad \text{MOS}_j = \frac{1}{M_j} \sum_{i=1}^{M_j} Z'_{ij}. \tag{4}$$

We illustrate the histogram of the MOSs of the LIEQ database in Figure 3(a) to observe the distribution of the MOSs. As illustrated in Figure 3(a), the perceptual quality spreads over the whole quality range. The majority of images have MOSs between 40 and 60, which indicates that the subjects are not satisfied with the quality of enhanced low-light images. To compare the enhanced effect of the LIEAs, we also illustrate the mean and standard deviation of the MOSs of the test images in Figure 3(b). It is observed that CRM and EFF perform the best among all methods, while

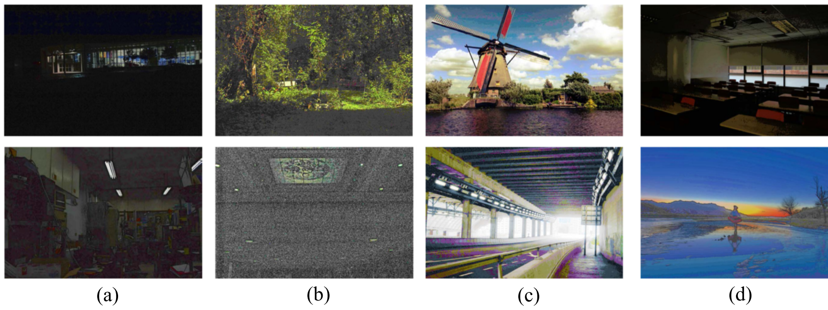


Fig. 4. Some typical distortion types in the LIEQ database: (a) low light, (b) noise, (c) color shift, (d) structural damage and over-enhancement.

HE and BPDHE have lower mean scores. We find the standard deviation of all LEAs are quite large, which indicates that the effectiveness of LIEAs is influenced by the image content.

3 OBJECTIVE QUALITY ASSESSMENT OF LOW-LIGHT IMAGE ENHANCEMENT

In this section, we study the low-light enhancement quality objectively and develop a novel **low-light image enhancement quality assessment (LIEQA)** model. We first observe the distortions and artifacts introduced in light enhancement. Some typical failures of enhanced images are shown in Figure 4. We find that low light effect, color shift, noise, structure damage, over-enhancement, and so on, are common distortions in light-enhanced images. A poorly enhanced image may have one or more of these distortions. Since distortions such as structure damage and over-enhancement destroy image structure, we evaluate the enhancement quality from four aspects: luminance enhancement, color rendition, noise, and structure preserving. For each aspect, we propose a metric to evaluate the corresponding enhancement effect. Finally, we integrate them into the final LIEQA measure.

3.1 Luminance Enhancement

The main goal of LIEAs is to lighten low-light images and recover the image content hidden in the dark area. Therefore, luminance enhancement is a very important aspect of LIEAs, and it directly determines how much content we can see in the enhanced image. Note that when an image is captured in an extremely low-light environment or at an extremely low exposure level, it is difficult or even impossible for LIEAs to recover the image content and increase the image luminance. So, it is necessary to evaluate the effect of luminance enhancement when evaluating the light enhancement quality.

The illumination map of an image reflects the ambient light condition when the image was shot. Illumination map estimation is also an important task for many LIEAs. Many of them utilize the optimization methods to estimate the illumination map based on the retinex model, such as alternating direction minimization technique [11, 13, 18]. But these methods are usually complex and time-consuming. Here, we do not need to estimate the exact illumination map, since we only want to calculate the global luminance enhancement effect. So, a rough illumination map of an image is estimated through computing the maximum of its three color channels, which is also frequently used in previous studies [12, 18]:

$$L(i,j) = \max_{c \in \{r,g,b\}} I^c(i,j), \tag{5}$$

where i and j are the spatial indices of the image, L is the illumination map.

Fig. 5. Some sample images with the same scene but different lighting conditions [77]. The EMD values of (a) to (e) are 15.19, 22.70, 31.05, 62.16, and 131.04, respectively.

Since local contrast normalization is often used to simulate the nonlinear masking of visual perception in many image processing applications [35], we adopt the similar model in Reference [47] to process the illumination map:

$$\tilde{L}(i,j) = \frac{L(i,j) - \mu}{\sigma + c}, \tag{6}$$

$$\mu(i,j) = \sum_{k,l} \omega_{k,l} I(i+k, j+l), \tag{7}$$

$$\sigma(i,j) = \sqrt{\sum_{k,l} \omega_{k,l} [I(i+k, j+l) - \mu(i,j)]^2}, \tag{8}$$

where μ is the local mean, and ω is a local Gaussian weighting window. Through Equations (5)–(8), we obtain the processed illumination maps of enhanced image and its corresponding reference image, which are denoted as $\tilde{L}_e$ and $\tilde{L}_r$.

A good luminance enhancement effect should make the illumination map of the enhanced image as close as possible to the illumination map of the reference image. If the luminance is increased excessively, then the enhanced image is over-exposure and loses local details. While if it is increased insufficiently, the enhanced image is still too dark to perceive the image content. Some similarity metrics like **mean square error (MSE)** fail to reflect the interaction of luminance between each pixel in the image. Here, we propose to use the **Earth Mover's Distance (EMD)** metric [60] as the luminance feature. EMD is a measure of the distance between two probability distributions and is commonly used in image retrieval filed as the luminance histogram feature. Previous studies [70, 71, 80] also indicate that EMD is an effective luminance indicator consistent with the subjective perception of human vision system. To intuitively show the effectiveness of using the EMD as the luminance feature, we illustrate some images with the same scene but different lighting conditions in Figure 5 and then calculate their EMD values between the illumination maps of these images and a black image. From Figure 5, it is seen that EMD values can effectively reflect the luminance feature of these images.

Since we want to evaluate the luminance similarity between the reference image and the enhanced image, we define the luminance feature as

$$l = \frac{\mathrm{EMD}(\tilde{L}_r, \tilde{L}_e)}{\mathrm{EMD}(\tilde{L}_r, \tilde{L}_{black})}, \tag{9}$$

where $\tilde{L}_r$, $\tilde{L}_e$, and $\tilde{L}_{black}$ are the histogram of the illumination map of the reference image, enhanced image, and a black image, respectively. EMD means the operator that calculates EMD value between two histograms. The reason we divide the $\mathrm{EMD}(\tilde{L}_r, \tilde{L}_e)$ with $\mathrm{EMD}(\tilde{L}_r, \tilde{L}_{black})$ is to eliminate the influence of image content, so we can fairly compare the luminance enhancement effect of images with different content luminance.

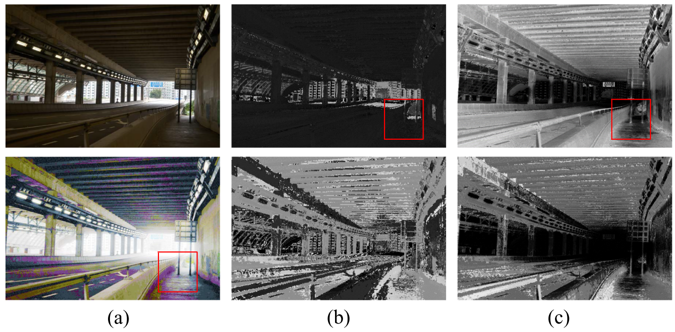


Fig. 6. An illustration for images with color shift in the hue and saturation map. We use the red rectangle to bound some color shift area. (a), (b), and (c) are the RBG images, hue maps, and saturation maps, respectively. Top: reference images, bottom: enhanced images.

According to the **contrast sensitivity function (CSF)** [1, 66], the luminance change users can perceive is proportional to the exponent of the luminance value. Therefore, the quality score of luminance enhancement is defined as

$$q_1 = e^{-\alpha l}, \tag{10}$$

where α is the parameter to control the importance of luminance enhancement.

3.2 Color Rendition

Color distortion is also one of the key factors that significantly affect image quality. From Figure 4, it is observed that LIEAs may introduce the color shift when recovering the original color information. Hence, we propose a metric to measure the color distortion of light-enhanced images.

The RGB color space is not intuitive for people to perceive the color information, so we convert the image I from RGB color space into HSV color space, which is more in line with the human vision system. We denote the hue, saturation, and value maps of the image I as H, S, and V, respectively.

Since hue and saturation represent the chrominance information and we have discussed the luminance information in Section 3.1, we just calculate the similarity of hue and saturation between the reference image and the enhanced image

$$C_H(i,j) = \frac{2H_r(i,j)H_e(i,j) + \epsilon_1}{H_r(i,j)^2 + H_e(i,j)^2 + \epsilon_1}, \quad C_S(i,j) = \frac{2S_r(i,j)S_e(i,j) + \epsilon_2}{S_r(i,j)^2 + S_e(i,j)^2 + \epsilon_2}, \tag{11}$$

where ϵ_1 and ϵ_2 are the small constants to avoid numerical instability and the ϵ in the remainder of this article has the same function.

From Figure 6, we notice that the color shift is more likely to occur in areas with lower hue and saturation values. Therefore, we introduce a weighting map to highlight the areas where the color shift occurs frequently. The weighting map is defined as

$$\boldsymbol{w}_c(i,j) = \frac{1}{H_r(i,j)S_r(i,j)}. \tag{12}$$

Therefore, the quality score of color rendition is defined as

$$q_2 = \frac{\sum_{i,j} \boldsymbol{w}_c(i,j)C_H(i,j)C_S(i,j)}{\sum_{i,j} \boldsymbol{w}_c(i,j)}. \tag{13}$$

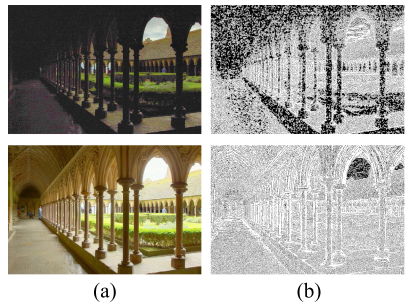


Fig. 7. Local binary pattern maps of the light-enhanced image and reference image. (a) and (b) are the RGB images and LBP maps, respectively. top: light-enhanced images, bottom: reference images.

3.3 Noise Evaluation

The noise is inevitable for low-light images, because the output of photosensitive sensors is buried by the intrinsic noise in the camera system if there is a lack of the sufficient amount of light. When LIEAs enhance the luminance component, they also amply the noise hiding in the dark area. Though it can alleviate the noise level through adopting denoising algorithms such as BM3D [8] to post-process the enhanced image or directly considering a noise term in the low-light enhancement model [29], the noise is still the main type of degradation for light-enhanced images. Here, we propose to use the **local binary pattern (LBP)** [53] to evaluate the noise degree in enhanced images.

LBP is a very simple but efficient visual descriptor that is frequently used in lots of image processing and computer vision applications. In References [30, 40, 76], LBP is also used to evaluate the image quality. For example, Wu et al. [76] propose an efficient blind IQA metric through selecting statistical features of local image structures extracted from LBPs. Li et al. [30] use the histograms of LBPs as one of the integrated quality features. Min et al. [40] utilize the LBPs to model the image sharpness and noise. These studies show that image degradation such as noise can change local structures modeled by LBPs.

LBP binarizes the difference between the values of a center pixel g_c and its circularly symmetric neighborhoods g_p. Then it adds the binarization results into a numerical value. Here, we use the LBP defined in Reference [40]:

$$\mathrm{LBP}_{\mathrm{P,R}} = \sum_{p=0}^{P-1} u(g_p - g_c), \tag{14}$$

where P, R are the neighbor number and radius of the LBP structure; $u(*)$ represents the unit step function

$$u(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}. \tag{15}$$

P and R are set 8 and 1, respectively, in this article, which denotes the simplest LBP structure. We empirically find that some values of LBPs are sensitive to the noise. Specifically, $\mathrm{LBP}_{8,1} \leq 6$ changes the most significantly. So, we further binarize the LBPs using the following formula:

$$\Gamma_{ij} = \begin{cases} 1 & \text{if } \mathrm{LBP}_{8,1} \leq 6 \\ 0 & \text{otherwise} \end{cases}. \tag{16}$$

We calculate the LBP maps of the reference image and the enhanced image using the above formulation and denote them as $\Gamma_r = (\gamma_{rij})_{h \times w}$ and $\Gamma_e = (\gamma_{eij})_{h \times w}$, respectively. As shown in Figure 7, LBP maps can effectively capture the structural change caused by noise. We then measure

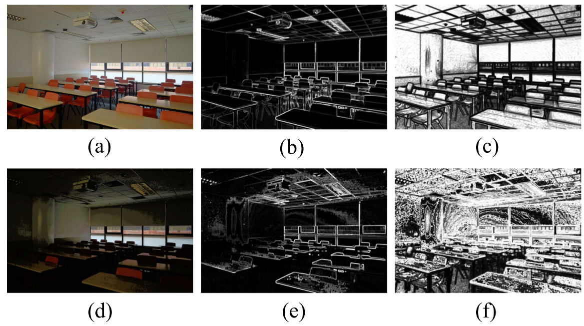


Fig. 8. An Illustration of the image with structure damage: (a) reference image, (b) variance map of the reference image, (c) weighting map, (d) enhanced image, (e) variance map of the enhanced image, (f) variance similarity map.

the similarity between Γ_r and Γ_e as the quality. First, the overlap and union between them are defined as

$$\begin{aligned} \Gamma_o &= (\gamma_{oij})_{h\times w} = (\gamma_{dij} \cdot \gamma_{mij})_{h\times w}, \\ \Gamma_u &= (\gamma_{uij})_{h\times w} = (\gamma_{dij} | \gamma_{mij})_{h\times w}. \end{aligned} \tag{17}$$

Then the quality score of noise can be defined as

$$q_3 = \frac{N_o}{N_u + 1}, \tag{18}$$

where N_o and N_u are the number of non-zero elements in the L_o and L_u maps, which can be calculated as

$$N_o = \sum_{i,j} \gamma_{oij}; \quad N_u = \sum_{i,j} \gamma_{uij}. \tag{19}$$

3.4 Structure Preserving

Structure information is extremely important for image quality assessment, since it can effectively capture image degradations. Note that some low-light images contain little image content information, which makes it difficult for LIEAs to recover the entire structural information. Thus, enhanced images will lose structural information. Some LIEAs fail to preserve the image naturalness, which can lead to image over-enhancement. Both of them cause structure distortions. We illustrate typical structural distortions in Figure 8 and Figure 9. Figure 8(d) shows the structure damage, where the structural information in many areas is lost. Figure 9(d) illustrates the over-enhancement, where some hardly observed image details are enhanced as image structures. In the following part, we propose two structure metrics to evaluate the two kinds of structural distortions.

3.4.1 Variance Similarity. Variance similarity has been widely used in many IQA methods to measure the image structure similarity [74, 75], since it is sensitive to structure change. Here, we use the variance structure similarity to capture the structure damage. Via Equations (7)–(8), we can derive the variance maps of the reference image and the enhanced image: $\boldsymbol{\sigma}_r$, $\boldsymbol{\sigma}_e$. Then, the variance structure similarity map $\boldsymbol{v}$ can be described by

$$\boldsymbol{v} = \frac{2\boldsymbol{\sigma}_r \cdot \boldsymbol{\sigma}_e + \epsilon_2}{\boldsymbol{\sigma}_r^2 + \boldsymbol{\sigma}_e^2 + \epsilon_2}. \tag{20}$$

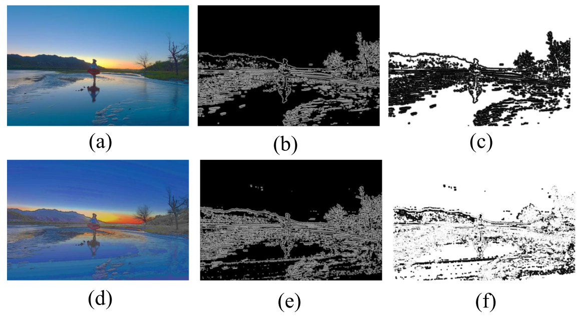


Fig. 9. An Illustration of the image with over-enhancement: (a) reference image, (b) edge variance map of the reference image, (c) weighting map, (d) enhanced image, (e) edge variance map of the enhanced image, (f) edge structure similarity map.

Figure 8 shows that smooth areas are more susceptible to structural damage; thus, we apply a content-based weighting map to highlight the smooth area:

$$\boldsymbol{w}_v = \frac{1}{\boldsymbol{\sigma}_r + \epsilon_3}. \tag{21}$$

The overall weighted structure damage then can be described as

$$q_s = \frac{\sum_{i,j} \boldsymbol{v}(i,j) \cdot \boldsymbol{w}_v(i,j)}{\sum_{i,j} \boldsymbol{w}_v(i,j)}. \tag{22}$$

3.4.2 Edge Structure Similarity. As illustrated in Figure 9(d), some low-contrast or near edge regions are often over-enhanced, which makes the edge areas thicker or new edges may appear in low contrast areas. Hence, we propose to use the edge structure similarity to capture the over-enhancement distortions. We apply the Log edge detector [50] on I_r and I_e to generate the binary edge maps $\boldsymbol{\zeta}_r$ and $\boldsymbol{\zeta}_e$. Then via Equations (7)–(8), we can derive the edge variance maps of the reference image and the enhanced image: $\boldsymbol{\rho}_r, \boldsymbol{\rho}_e$. So the edge structure similarity map is calculated as

$$\boldsymbol{e} = \frac{2\boldsymbol{\rho}_r \cdot \boldsymbol{\rho}_e + \epsilon_4}{\boldsymbol{\rho}_r^2 + \boldsymbol{\rho}_e^2 + \epsilon_4}. \tag{23}$$

Similarly, it is observed that the over-enhancement often occurs in the non-edge areas in Figure 9. So, the reciprocal of the variance of the edge map is introduced as the weighting map to highlight the areas where over-enhancement occurs frequently:

$$\boldsymbol{w}_e = \frac{1}{\boldsymbol{\rho}_r + \epsilon_5}. \tag{24}$$

Then the overall over-enhancement is computed as

$$q_o = \frac{\sum_{i,j} \boldsymbol{e}(i,j) \cdot \boldsymbol{w}_e(i,j)}{\sum_{i,j} \boldsymbol{w}_e(i,j)}. \tag{25}$$

We incorporate the variance similarity and edge structure similarity to the final structure-preserving score

$$q_4 = q_s \cdot q_o^{\beta}, \tag{26}$$

where β represents the relative importance between structural damage and over-enhancement.

3.5 Overall Quality Evaluation

Finally, we evaluate the overall quality of light-enhanced images by multiplying scores of luminance enhancement, color rendition, noise evaluation, and structure preserving:

$$q = q_1 \cdot q_2 \cdot q_3 \cdot q_4. \tag{27}$$

Besides the above overall quality term, the four components can be also used for light enhancement quality evaluation from a specific aspect.

4 EXPERIMENTAL VALIDATION

4.1 Experiment Protocol

The proposed method is validated on the constructed LIEQ database. To demonstrate the effectiveness of the proposed LIEQA index, we compared it with the state-of-the-art FR IQA models including PSNR, SSIM [74], MS-SSIM [75], FSIM [88], GMSD [78], GSI [32], VSI [87], MAD [27], TMQI [81], FSITM [51], and DEHAZEfr [43], where TMQI and FSITM are designed for tone-mapping IQA, DEHAZEfr is proposed for dehazing IQA, and others are general-purpose FR IQA methods. In addition, frequently used evaluation metrics in LIEA studies are tested on the LIEQ database, which includes **lightness order error (LOE)** [73], color constancy [64], contrast gain [11], discrete entropy [64], BTMQI [16], BRISQUE [48], NIQMC [15], NIQE [49], and NRERM [17]. Since the distortion types of tone-mapped images overlap with that of light-enhanced images to some extent, we also test a popular NR tone-mapping IQA method HRGRADE [26] on the LIEQ database.

Before evaluating the performance of IQA models, we follow the common procedures in Reference [62] to map the scores predicted by IQA models using a five-parameter logistic function that is formulated as

$$q(s) = \tau_1 \left(\frac{1}{2} - \frac{1}{1 + e^{\tau_2 (s - \tau_3)}} \right) + \tau_4 s + \tau_5, \tag{28}$$

where $\{\tau_i | i = 1, 2, \ldots, 5\}$ are parameters that need to be fitted, and s and $q(s)$ represent the predicted scores and mapped scores, respectively.

Then four consistency evaluation criteria are used to compare the correlation between the mapped scores and MOSs. They are **Spearman Rank Correlation Coefficient (SRCC), Kendall's rank order correlation coefficient (KRCC), Pearson Linear Correlation Coefficient (PLCC)**, and **Root Mean Squared Error (RMSE)**, respectively. These four statistical indexes describe different aspects for evaluating the performance of IQA models. To be more specific, SRCC and KRCC both reflect the prediction monotonicity. PLCC and RMSE reflect prediction linearity and prediction accuracy, respectively. Note that an excellent model should obtain values of SRCC, KRCC, and PLCC close to 1, and the value of RMSE near 0.

4.2 Performance Comparison with FR IQA Models

The performances of the LIEQA and 11 state-of-the-art FR IQA models are listed in Table 1. From Table 1, we have several useful observations. First, the proposed method achieves the best performance among all compared models and it leads by a large margin, which demonstrates its effectiveness for low-light image enhancement quality evaluation. Then, we observe that state-of-the-art traditional FR IQA models do not have enough ability to predict the quality of light-enhanced images. Some previous LIEA studies [12, 64, 72] use the prevailing FR IQA methods such as PSNR and SSIM to evaluate the enhancement effect, however, the results may not be accurate. The reason is that most of these FR IQA models are developed and validated on the traditional IQA databases, where the distortion type of each image is single and artificially introduced and simulated. In comparison, light-enhanced images usually contain multiple types of distortions and there is no

Table 1. Performance of 11 State-of-art FR IQA Models and the Proposed Metric on the LIEQ Database

Critera \ Metrics	A PSNR	B SSIM	C MS-SSIM	D FSIM	E GMSD	F GSI	G VSI	H MAD	I TMQI	J FSITM	K DEHAZEfr	L Proposed
SRCC	0.3355	0.5826	0.6398	0.6669	0.6479	0.5585	0.6700	0.7068	0.5373	0.6137	0.7378	**0.8411**
KRCC	0.2283	0.4038	0.4524	0.4762	0.4591	0.3862	0.4771	0.5080	0.3861	0.4378	0.5382	**0.6407**
PLCC	0.3423	0.5880	0.6505	0.6786	0.6677	0.5681	0.6799	0.7061	0.5510	0.6247	0.7400	**0.8390**
RMSE	13.2933	11.4435	10.7452	10.392	10.5327	11.6430	10.3745	10.0181	11.8070	11.0474	9.5154	**7.6981**

We highlight the best-performing model in each row.

specific law for each distortion type. The proposed method considers this phenomenon and evaluates the light-enhanced images from different aspects and thereby significantly improves the evaluation performance. Finally, we compare the performances of FR IQA methods designed for enhanced images. The performances of TMQI and FSITM are similar to traditional FR IQA models and are both inferior to the proposed LIEQA index. The reason is that existing tone-mapping IQA methods emphasize image structure and naturalness. However, distortion types such as noise and color shift have an important impact on the visual quality of light-enhanced images. Therefore, tone-mapping IQA methods may evaluate the quality of low-light enhanced images on some aspects like image structure, but still cannot assess the overall quality well. DEHAZEfr achieves a relatively good result on the LIEQ database because dehazing algorithms and low-light enhancement algorithms both try to recover the image content hidden by haze or darkness. So, dehazed images and light-enhanced images have some distortions in common, such as structure damage, over-enhancement, and so on. But dehazed images do not include significant luminance change and noise, so IQA metrics for dehazed images are also insufficient for evaluating light-enhanced images.

We further conduct statistical significance tests to verify whether the performance of the proposed method is statistically better than others. The conducted statistical test is similar to the one introduced in Reference [62]. We compare the variances of residuals between the nonlinear mapped scores and subjective ratings. Note that higher variance means worse performance and vice versa. The utilized F-statistic is based on the ratio of two model's residual variances. The null hypothesis is set as the residuals of two quality metrics derive from the same distribution and they are statistically indistinguishable with a 95% confidence. We test all possible pairs of models and record the significance test results in Figure 10. We can find that the proposed model is significantly superior to others.

4.3 Performance Comparison with NR Quality Models and Descriptors

Besides FR IQA models, previous LIEAs studies also used some NR IQA measures or descriptors to quantitatively evaluate the effect of image enhancement. In this section, we test these metrics on the LIEQ database to verify their effectiveness and compare them with the proposed method. Generally, we can divide these metrics into two categories.

- NR IQA methods: BTMQI [16], NIQMC [15], NIQE [49], NRERM [17], BRISQUE [48], and HRGRADE[26]. BTMQI and NIQMC are NR IQA measures for contrast-enhanced images, HRGRADE is a NR IQA measure for tone-mapped images, and NIQE, NRERM, BRISQUE are general-purpose NR IQA measures.
- Quality descriptors for light enhancement: **Lightness order error (LOE)** [73], color constancy [64], contrast gain [11], discrete entropy [64]. Each of them reflects one aspect of light enhancement quality, such as luminance, color, contrast, and so on. These descriptors have been used for low-light enhancement quality evaluation in previous studies.

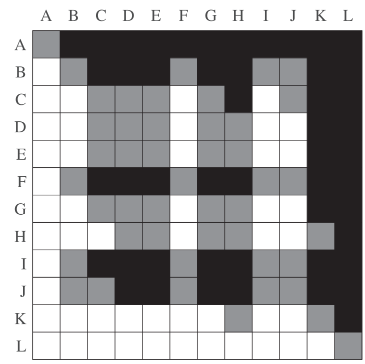


Fig. 10. Statistical test results of the proposed method and compared FR IQA methods on the LIEQ database. Note that a black/white block means the row method is statistically worse/better than the column one. A gray block means the row method and the column method are statistically indistinguishable. The methods denoted by A–L are the same as the ones in Table 1.

Table 2. Performance Comparison with NR IQA Models and Descriptors

Critera \ Metrics	LOE	Color Constancy	Contrast Gain	Discrete Entropy	BTMQI	NIQMC	BRISQUE	NIQE	NRERM	HIGRADE	Proposed
SRCC	0.1372	0.5004	0.1321	0.4680	0.4351	0.3046	0.4906	0.2480	0.1725	0.6067	**0.8411**
KRCC	0.0913	0.3486	0.1449	0.3210	0.2917	0.2228	0.3403	0.1775	0.2614	0.4318	**0.6407**
PLCC	0.1662	0.5136	0.2430	0.4913	0.4323	0.3556	0.5553	0.3207	0.4055	0.6326	**0.8390**
RMSE	13.9514	12.1392	13.7241	12.3229	12.7575	13.2233	11.7668	13.4009	12.9330	10.9574	**7.6981**

We list the performances of these measures in Table 2. From Table 2, we first observe that none of them is effective enough to evaluate the quality of low-light enhancement. It is easy to understand these results, since we have shown that light-enhanced images contain more complex distortions than the distortions considered in traditional IQA studies. Therefore, general-purpose NR IQA methods developed and verified on traditional IQA databases fail to evaluate the quality of light-enhanced images. NR IQA methods for contrast-enhanced images and descriptors designed for light enhancement only consider one aspect of the image distortions, so they also can not evaluate the overall image quality comprehensively. Then, we find that HIGRADE, designed for tone-mapped images, achieve the best performance among all NR IQA methods and descriptors, and also has comparative performance with the FR tone-mapping IQA methods, which indicates that the features extracted for NR tone-mapping IQA may also work for NR IQA for light-enhanced images. Finally, comparing Table 1 and Table 2, we can observe that FR IQA models are generally more effective than NR IQA models, which is not surprising. This is also the reason why we introduce the well-exposed image as the reference and propose the low-light enhancement quality measure following a FR IQA framework.

4.4 Ablation Experiment: Feature Analysis

As described in Section 3, the proposed method consists of several key components, which include luminance enhancement, color rendition, noise evaluation, and structure preserving. In this section, we conduct a series of ablation experiments to analyze the contribution of each component. First, we test the proposed model without luminance enhancement, color rendition, noise evaluation, and structure-preserving components, respectively, on the LIEQ database to verify their contributions to the overall image quality prediction. Then, we validate these four components

Table 3. Performance Results of the Ablation Experiments

Case	q1	q2	q3	q4	SRCC	KLCC	PLCC	RMSE
1	√	√	√	√	0.8411	0.6407	0.8390	7.6981
2	×	√	√	√	0.8383	0.6371	0.8374	7.7325
3	√	×	√	√	0.8169	0.6153	0.8152	8.1946
4	√	√	×	√	0.8219	0.6203	0.8210	8.0769
5	√	√	√	×	0.7770	0.5696	0.7803	8.8475
6	√	×	×	×	0.4164	0.2830	0.5947	12.7753
7	×	√	×	×	0.6898	0.4917	0.7858	10.0332
7	×	×	√	×	0.5939	0.4161	0.7051	11.3742
9	×	×	×	√	0.7809	0.5816	0.8058	8.7503

q1: luminance enhancement, q2: color rendition, q3: noise evaluation, q4: structure preserving.

individually on the LIEQ database to show their effectiveness in evaluating the image quality from specific aspects.

The results of ablation experiments are listed in Table 3. It is observed that the performances of *Case 2* to *Case 5* are all inferior to the complete algorithm, which indicates that all components make contributions to the overall algorithm. We find that *Case 5* has the lowest performance, which means that image structure is the most important for quality assessment of light-enhanced images. Besides the structure preserving, the color rendition also makes a great contribution to the quality evaluation of light-enhanced images. Luminance enhancement and noise evaluation terms contribute relatively little to the overall evaluation when compared to others. The performances of *Case 6* to *Case 9* also validate this point. In addition, we notice that the performances of the noise-evaluation component, the structure-preserving component, the luminance-enhancement component, and the color-rendition component are better than PSNR, SSIM, LOE descriptor, and Color Constancy descriptor, respectively. Note that PSNR, SSIM, LOE descriptor, and Color Constancy descriptor are frequently used in the literature to evaluate noise, image structure, luminance, and color, respectively. Therefore, these four components can also be used for light-enhancement quality evaluation from a specific aspect.

4.5 Performance Comparison of Using Various Edge Detectors

The proposed method adopts the Log edge detector to generate the edge map. For comparison, we also test the performance of some other commonly used edge detectors, including Roberts [59], Sobel [21], Prewitt [7], and Canny [5]. The performance of each edge detector is listed in Table 4. It can be observed that the Log edge detector achieves the best performance. This may attribute that Log edge detector is more robust to the noise and more likely to detect true weak edges. The performance of using any edge detector is better than the proposed method without incorporating the edge structure similarity term, which also indicates the importance of edge similarity to the overall low-light image enhancement quality assessment.

4.6 Parameter Sensitivity

The proposed method involves two main parameters: α in Equation (10) and β in Equation (26), which are used to determine the importance of luminance enhancement and over-enhancement, respectively. In this article, we empirically set the values of α and β as 2.2 and 3.5, respectively. Here, we vary the values of α and β to test the model stability to parameters. To be specific, the

Table 4. Performance Results of Different Edge Detectors

Critera	Soblel	Prewitt	Roberts	Canny	Log (pro.)
SRCC	0.8229	0.8227	0.8168	0.8352	0.8411
KLCC	0.6227	0.6224	0.6148	0.6334	0.6407
PLCC	0.8236	0.8234	0.8175	0.8322	0.8390
RMSE	8.0235	8.0276	8.1479	7.8446	7.6981

Table 5. SRCC Performance Results of the Parameter Sensitivity Tests

β \ α	2.0	2.1	2.2	2.3	2.4	2.5
2.5	0.8394	0.8394	0.8394	0.8394	0.8394	0.8393
3.0	0.8407	0.8406	0.8407	0.8407	0.8407	0.8406
3.5	0.8409	0.8410	0.8411	0.8410	0.8410	0.8410
4.0	0.8405	0.8405	0.8405	0.8405	0.8405	0.8405
4.5	0.8393	0.8394	0.8394	0.8394	0.8394	0.8394

parameter α is set from 1.0 to 3.0 with a step of 0.5 and the parameter β is set from 2.0 to 2.5 with a step of 0.1. We change both parameters at the same time. The performance in terms of SRCC is listed in Table 5. From Table 5, we can obtain that the performance of the proposed method remains quite stable over a pretty large range of test parameters.

4.7 Computational Complexity

The computational complexity of the model is important for its application prospects. In this section, we discuss the computational complexity of the proposed LIEQA model and other IQA models. The source codes of the compared IQA methods are provided by the authors. Specifically, we run these codes on an image with a fixed resolution of 512×512 for 1,000 times on a computer with 4.00 GHz Intel Core i7-6700K CPU, 32 GB RAM, and report the average running time in Table 6. From Table 6, we can observe that the proposed LIEQA model has a moderate computational complexity compared to other IQA methods. Though the computational complexities of models such as PSNR, SSIM, GMSD, and so on, are very low, their performances on evaluating the quality of light-enhanced images are significantly inferior to the proposed model. Overall, the proposed model achieves a good tradeoff between running time and performance.

4.8 Discussion

As described in Section 4.2 and Section 4.3, FR quality measures are generally more effective than NR quality models. Considering this, we introduce the multi-exposure images and use the well-fused image as the reference. The difficulty of evaluating low-light enhancement quality is largely reduced by doing so. The proposed measure can act as an evaluator for various low-light enhancement algorithms and systems. For algorithm or system testing, we can always find some multi-exposure images and the corresponding well-fused ones. While in some other applications where the input is not determinable or available, the proposed method is not applicable. NR quality measures are needed in these situations, and we will look into this problem in our future work.

Table 6. Comparison of Computational Complexity for 11 State-of-the-art FR IQA Models and the Proposed LIEQA Model

Metrics	PSNR	SSIM	MS-SSIM	FSIM	GMSD	GSI	VSI	MAD	TMQI	FSITM	DEHAZEfr	Proposed
Time	0.0010	0.0154	0.0491	0.2499	0.0050	0.0228	0.1216	1.5268	0.1434	0.4183	0.1045	0.4950

Time: seconds/image.

5 CONCLUSION

In this article, we investigate the quality assessment problem of low-light image enhancement comprehensively. We first construct a low-light image enhancement quality (LIEQ) database and conduct a large-scale subjective quality assessment study on the LIEQ database. This subjective study, on one hand, provides a benchmark of state-of-the-art LIEAs, and on the other hand, facilitates the objective quality evaluation study. The LIEQ database consists of 1,000 light-enhanced images generated from 100 low-light images through 10 low-light image enhancement algorithms. The corresponding reference images derived via multi-exposure fusion and stack-based HDR methods are also provided in the database, which makes it suitable for both FR and NR light enhancement quality evaluation studies.

It is observed that light-enhanced images are generally degraded by complex distortions that are significantly different from traditional distortions, and state-of-the-art IQA models are not suitable for evaluating the quality of low-light enhancement. Therefore, we develop a novel FR low-light image enhancement quality assessment (LIEQA) model. The LIEQA index evaluates the enhancement quality from four different perspectives: luminance enhancement, color rendition, noise evaluation, and structure preserving, which are described via histograms' earth mover's distance, chrominance similarity, LBP similarity, and variance and edge structure similarity, respectively. Experimental results on the LIEQ database show that the proposed method is significantly superior to state-of-the-art general-purpose or light enhancement–specific quality models. The LIEQ database is the largest of its kind so far, while the LIEQA index is the first comprehensive low-light enhancement quality evaluator. Both of them can promote the low-light enhancement and evaluation research.

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